ZCAS UNIVERSITY

SCHOOL OF COMPUTING, TECHNOLOGY & APPLIED SCIENCES

CCS5421 INTERNET OF THINGS

END OF SEMESTER EXAMINATION

MAY/JUNE 2026

TIME:

TIME ALLOWED : WRITING – THREE HOURS

READING – 5 MINUTES

INSTRUCTIONS:

1. Section A: This question is **compulsory** and must be attempted.
2. Sections B: Answer **TWO (2)** questions from this section.
3. This examination paper carries a total of **100 marks.**
4. Candidates must **not turn this page** until the invigilator tells them to do so.

SECTION A: Question 1 is compulsory and must be attempted

Question 1

A local authority in Zambia plans to implement a **Smart Urban Services Platform** to improve service delivery. The proposed platform will include smart parking, intelligent traffic lights, smart water management, environmental monitoring, and a public dashboard for citizens and decision-makers.

The system will use sensors, smart meters, cameras, GPS devices, gateways, cloud platforms, mobile applications, dashboards, and data analytics. Some components will require real-time response, while others will only require periodic reporting. The local authority is also concerned about unreliable connectivity, high data volumes, privacy risks, cybersecurity threats, vendor lock-in, and the cost of maintaining the system.

Using the above scenario, answer the following questions:

a) Design a high-level IoT architecture for the proposed Smart Urban Services Platform. Your answer should identify the major components, including sensors, actuators, controllers, gateways, connectivity, cloud/backend platform, application layer, user interface, and security layer. **[10 marks]**

b) Select suitable communication and application-layer standards for at least THREE services in the scenario, such as smart parking, traffic lights, water management, or environmental monitoring. Justify your choices using factors such as range, bandwidth, latency, power consumption, cost, reliability, and scalability. **[10 marks]**

c) Analyse the big data challenges likely to arise from the platform. Your answer should discuss volume, velocity, variety, veracity, value, storage, processing, data quality, integration, and data governance. **[10 marks]**

d) Evaluate the privacy and cybersecurity risks associated with the proposed system. Your answer should include location tracking, number plate data, household water consumption patterns, video data, unauthorised access, weak authentication, insecure APIs, and data ownership. **[10 marks]**

e) Some services, such as intelligent traffic lights and water monitoring, may become mission critical. Discuss the reliability, availability, safety, accountability, and human oversight requirements for such systems. Propose practical measures to ensure resilience. **[10 marks]**

(50 Marks)

SECTION B: Attempt any TWO (2) questions in this section

Question 2

IoT standards are essential because IoT devices are produced by different manufacturers and must communicate reliably across different systems, networks, and platforms. The lecture material highlights that IoT standards support interoperability, scalability, security, reliability, cost reduction, and global adoption.

a) Explain why IoT cannot grow effectively without standards. **[6 marks]**

b) Compare any FIVE IoT communication standards from the following: Wi-Fi, Bluetooth Low Energy, Zigbee, Z-Wave, LoRaWAN, NB-IoT, and 5G. Your answer should consider range, power use, bandwidth, strengths, limitations, and typical use cases. **[10 marks]**

c) Discuss the role of networking standards such as IPv6, 6LoWPAN, and RPL in IoT systems. **[9 marks]**

(25 marks)

Question 3

IoT is increasingly embedded in everyday life through smart homes, healthcare devices, transport systems, agriculture, utilities, classrooms, security systems, and smart cities. It also raises important questions about privacy, data governance, cybersecurity, affordability, and inclusion.

a) Explain the relevance of IoT for the future of society, public services, business, agriculture, transport, healthcare, and utilities. **[5 marks]**
b) Analyse TWO of the following IoT applications using the analytical framework of problem addressed, IoT components, data collected, benefits, risks, and governance implications **[10 marks]:**
   i. Lighting as a Service;
   ii. Intelligent Traffic Systems;
   iii. Smart Parking;
   iv. Smart Water Management.
c) Discuss the three major implementation challenges in IoT for big data management and connectivity challenges. **[6 marks]**
d) Differentiate between the Internet of Things and the Internet of Everything. Use examples to show how people, processes, data, and things are connected. **[4 marks]**

(25 marks)

Question 4

IoT applications are increasingly relevant in both global and local contexts, including agriculture, environmental monitoring, health, transport, smart campuses, and utility systems. A complete IoT system typically includes sensing devices, communication links, processing, storage, and decision-making components. However, designers must also decide where data processing should occur. Some systems send data to the cloud for storage and analysis, while others process part of the data locally at the edge or gateway to reduce delay, bandwidth use, and dependence on constant internet connectivity.

a) Identify and discuss IoT applications that are most relevant to Zambia or your local community. **[8 marks]**
b) Can an IoT system work effectively if one major component is missing? Explain your answer with examples. **[9 marks]**
c) Should IoT data always be sent to the cloud, or should some processing happen locally? Discuss. **[8 marks]**

(25 marks)

END OF EXAMINATION